ETHICAL AND LEGAL CHALLENGES IN THE DEVELOPMENT OF AUTONOMOUS CARS

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OUTLINE

1. Features of autonomous cars
2. Ethical challenges
3. Legal challenges

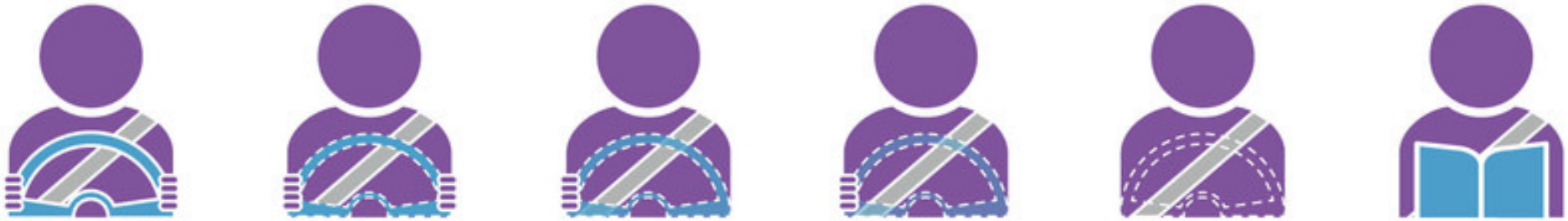
THE ROAD SAFETY ISSUE

- Approximately 1.3 million people die each year as a result of road traffic crashes.
- Road traffic crashes cost most countries 3% of their gross domestic product.
- Road traffic injuries are the leading cause of death for children and young adults aged 5-29 years.

(Source: WHO)

SYNOPSYS®

LEVELS OF DRIVING AUTOMATION



0	1	2	3	4	5
NO AUTOMATION	**DRIVER ASSISTANCE**	**PARTIAL AUTOMATION**	**CONDITIONAL AUTOMATION**	**HIGH AUTOMATION**	**FULL AUTOMATION**
Manual control. The human performs all driving tasks (steering, acceleration, braking, etc.).	The vehicle features a single automated system (e.g. it monitors speed through cruise control).	ADAS. The vehicle can perform steering and acceleration. The human still monitors all tasks and can take control at any time.	Environmental detection capabilities. The vehicle can perform most driving tasks, but human override is still required.	The vehicle performs all driving tasks under specific circumstances. Geofencing is required. Human override is still an option.	The vehicle performs all driving tasks under all conditions. Zero human attention or interaction is required.

THE HUMAN MONITORS THE DRIVING ENVIRONMENT

THE AUTOMATED SYSTEM MONITORS THE DRIVING ENVIRONMENT

DEFINITIONS

- Automated vehicles: car has systems that automate certain functions, such as adaptive cruise control (which maintains a certain distance from the vehicle in front of the car) and automatic braking or, at the driver's direction, takes over driving subject to the driver retaking control if necessary.
- Autonomous vehicles: highly automated car that drives itself in most or all conditions (level 3-5)
- Connected autonomous vehicles: car can access the internet and is usually connected to a wireless network, and it can communicate with other vehicles, traffic management infrastructure, manufacturers, and fleet operators etc.

FEATURES OF AUTONOMOUS CARS

- Driverless: drive themselves with little or no human intervention
- Machine-made decisions: built-in sensing systems and artificial intelligence (AI) programs to navigate the roads
- Connected navigation: GPS, data collection and sharing
- Road safety: eliminate the human drive error, effective detection of and responses to objects and obstacles, reducing the epidemic of traffic incidents and fatalities
- Other benefits: reduce emissions and ease congestion

Self-Driving vs Regular Cars

Comparison Chart

Regular Cars	Self-Driving Cars
Human-driven cars monitored by humans sitting behind the steering wheel.	Self-driving cars are vehicles that drive themselves with little or no human intervention.
Driver has the control over driving but involves some kind of automation.	Use their own sensing systems and software programs to navigate the roads.
Human driven cars account for more road fatalities because of human drive error.	Designed to eliminate human drive error, reducing the epidemic of traffic incidents and fatalities.
Basic levels of automation used are electronic stability control (ESC), blind spot detection, antilock brake systems (ABS), etc	Use sensors and software programs, along with AI to help the car navigate the roads and avoid obstacles.

ETHICAL CHALLENGES

1. Personhood argument
2. Ethical dilemmas
3. Solutions?

IS AUTONOMOUS CAR A PERSON?

- AI: the ability of a computer (or a robot/machine controlled by a computer) to do tasks that are usually done by humans because they require human intelligence and discernment.

- AI: able to mimic the problem-solving and decision-making capabilities of the human mind.
- "Consider what person stands for; which, I think, is a thinking, intelligent being, that has reason and reflection." (per The English philosopher John Locke)

CONT'D

- No consciousness: they are machines that are not aware of themselves; cannot perceive their internal states; have no understanding of other people's emotions or behaviours

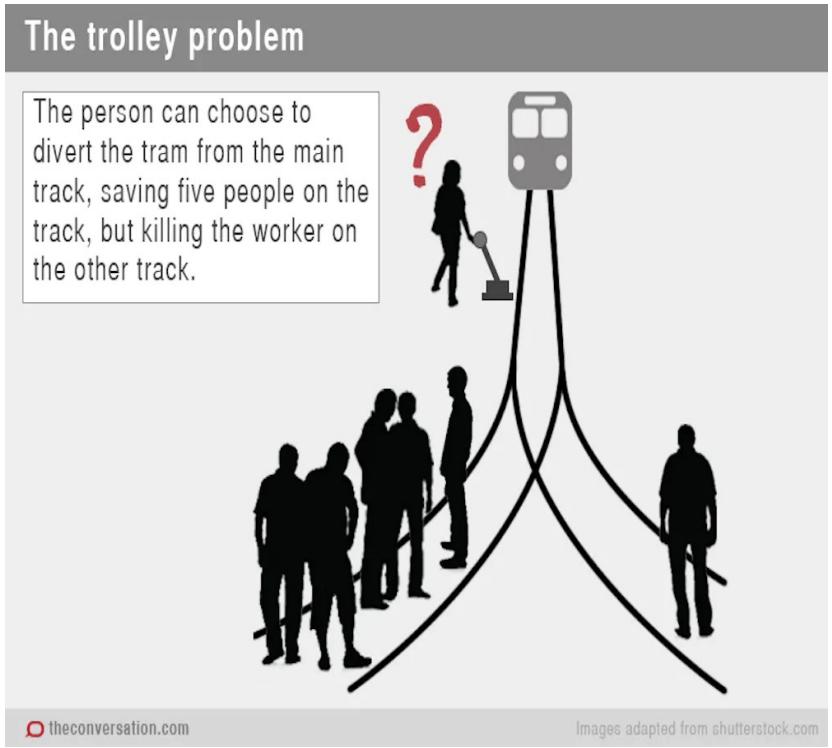
- No free will: they cannot step outside of the system and decide against the algorithms or the data it has been given; programmed with algorithms to follow;

- Here arise another ethical issue – how humans should determine the moral and legal algorithms for AI to "act"?

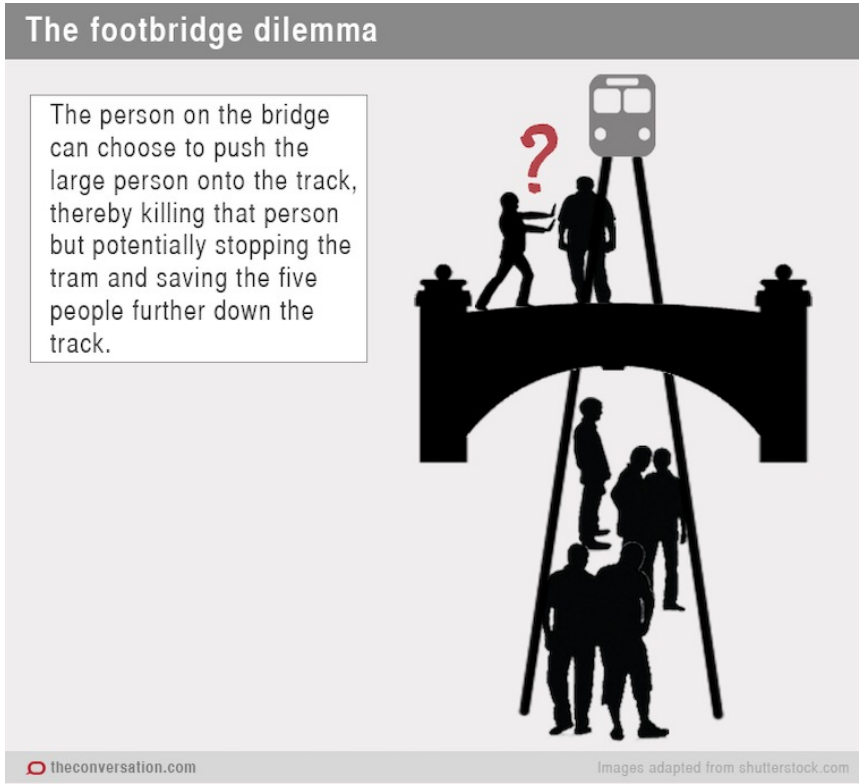
ETHICAL CHALLENGES

1. Personhood argument
2. Ethical dilemmas
3. Human autonomy

TROLLEY PROBLEM



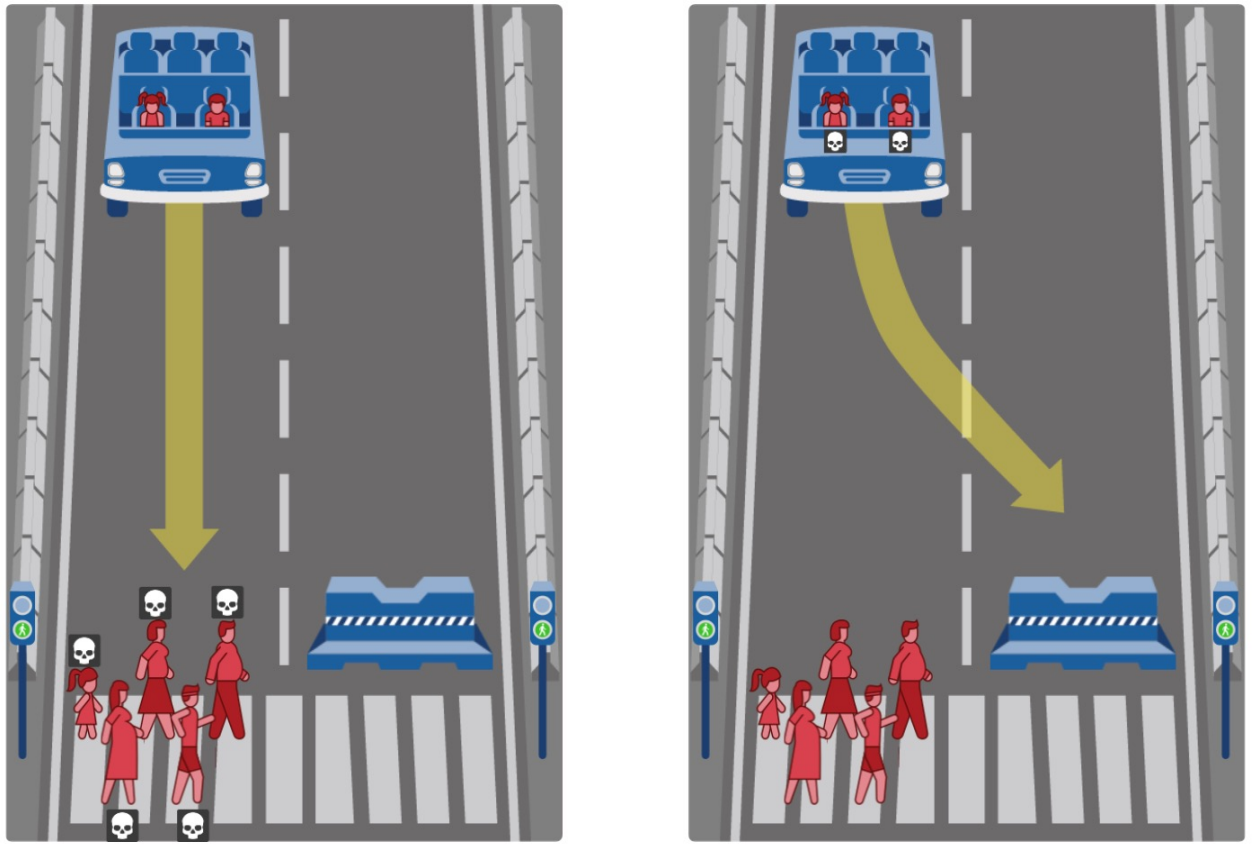
THE FOOTBRIDGE DILEMMA



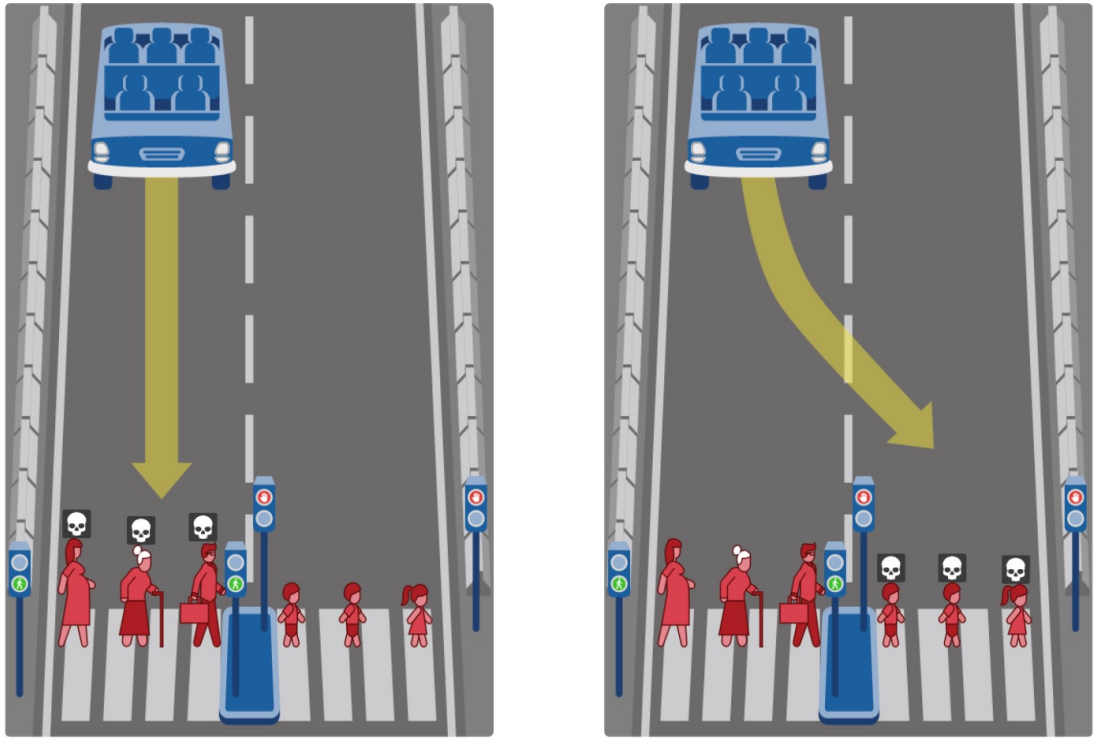
BRAKE FAILURE SCENARIOS

- Human first, animal second, and things third?
- Save more people?
- The young v. the elderly?
- The family v. the stranger?
- Inside v. outside the car?
- Save people who comply with laws?
- Save people who contribute more to the society?

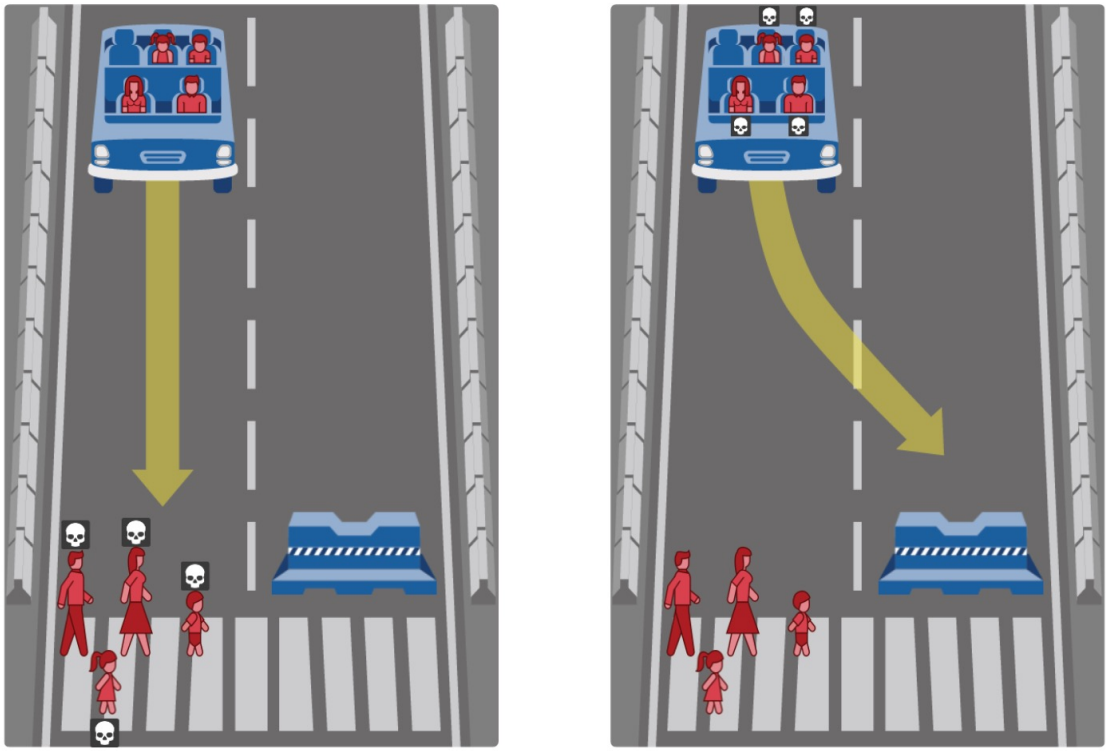
MORE V. LESS



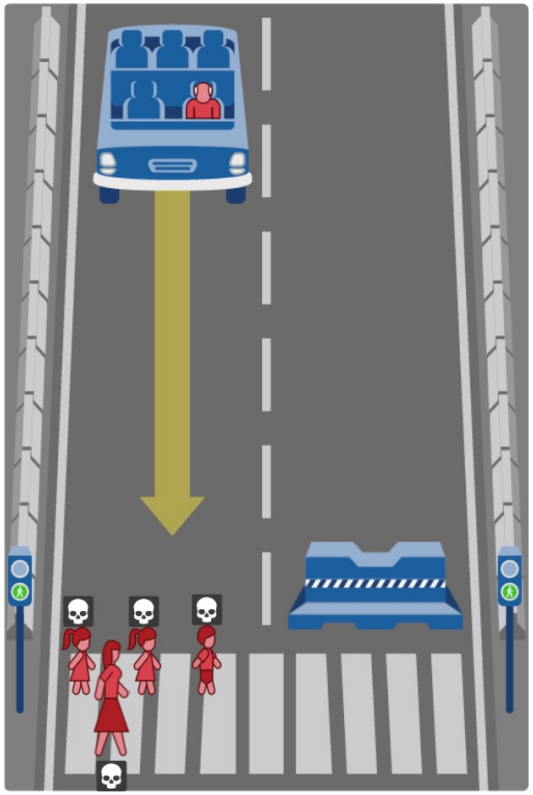
YOUNG V. ADULT (ELDERLY)

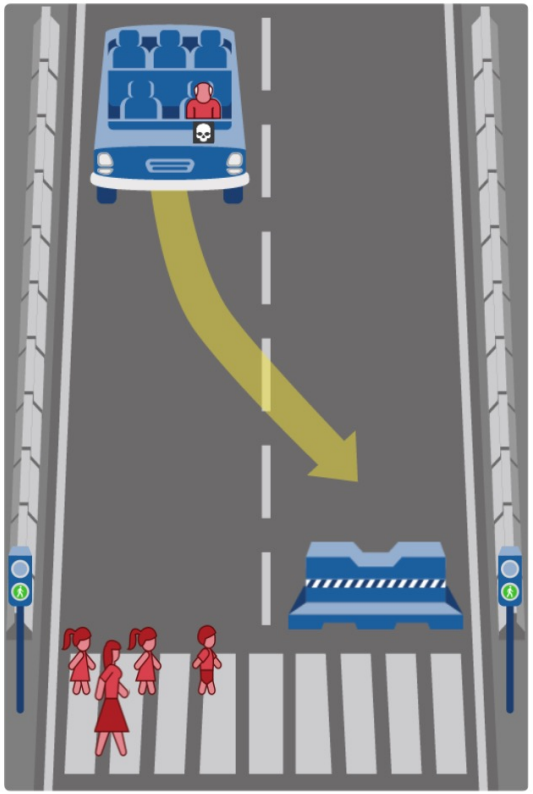


FAMILY V. STRANGER

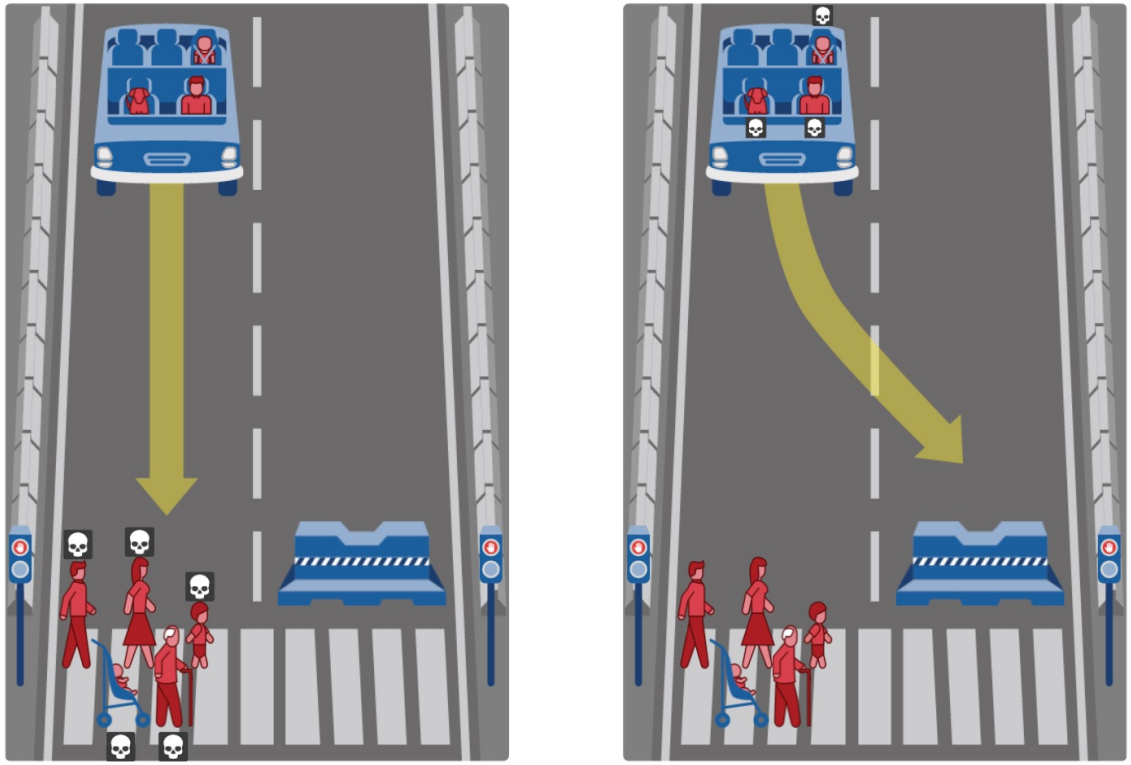


INSIDE V. OUTSIDE

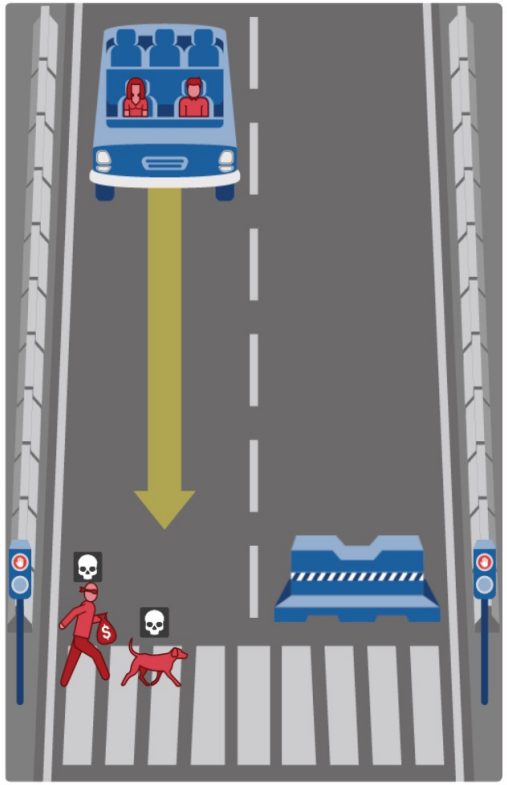


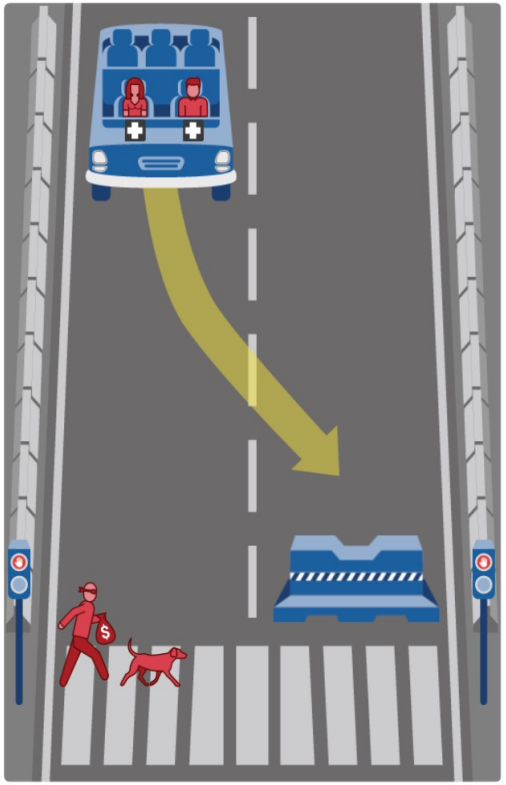


COMPLYING WITH THE LAW?



BAD PERSON & ANIMAL





VOTE FOR A MAJORITY ETHICAL PREFERENCE?

- The Moral Machine: three main discoveries so far (Awad, et al., The Moral Machine Experiment, *Nature* 563 (2018), 59-64)
  1. People prefer to spare humans over animals;
  2. People prefer to spare more lives over fewer lives;
  3. People prefer to spare young lives over old lives
- Preference-satisfaction consequentialism:
  1. The right act is the one that brings about the good consequence;
  2. The good consequence is the one that satisfies more preferences than any alternative.

FUNDAMENTAL ETHICAL THEORIES

- With what rules should we programme autonomous cars to handle ethical dilemmas to be encountered?
    - Deontological theory
    - Utilitarian theory
    - Confucian virtue ethics

DEONTOLOGICAL THEORY

- Asimov's Three Laws were first introduced in Asimov's science fiction story *Runaround* in 1942.

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.

Q: Can Asimov's Three Laws solve the trolley problem and the footbridge dilemma?

UTILITARIAN THEORY

- An act is right if and only if it produces a greater balance of good over bad in its consequences for all people affected, than any other act available to the agent.
- In weighing the consequences of available courses of action, autonomous cars should be programmed to choose the option that will either generate the maximum benefits, or prevent the greater harm, for all human beings involved.

SOCIAL UTILITY V. SELF-INTEREST

- In a joint study conducted by Jean-François Bennefon of the University of Toulouse, Azim Shariff of the University of Oregon, and Iyad Rahwan of MIT (2016):

  Q1: Do you favor sacrificing yourself and the driver inside an autonomous car to avoid running over a group of pedestrians?

  Q2: Would you purchase such a utilitarian autonomous car?

- The researchers noted that ”even though participants still agreed that utilitarian autonomous cars were the most moral, they preferred the self-protective model for themselves.” (Bennefon et al 2016)
- Moral machine project

UTILITARIAN OR SELFISH AUTONOMOUS CARS?

- Humans would refrain from certain actions that involve clear personal harm.
- Humans do not always favor a utilitarian moral consideration, especially when the smaller number includes themselves, their kin or their acquaintances.
- Except for exceptional heroic acts, few people would willingly sacrifice themselves or their loved ones in order to produce the greater utility for all.

CONFUCIAN VIRTUE ETHICS

- Among many highly emphasized virtues in the Analects that can be formulated into moral principles/rules for Confucian autonomous cars (per JeeLoo Liu)
    - loyalty (zhong 忠),
    - reciprocity (shu 恕),
    - humanity (ren 仁).
- “Loyalty” is directed towards the role one plays. In this sense, it can be defined as “doing what one is supposed to do” or “being loyal to one’s role.”
- Rule 1: an autonomous car must first and foremost fulfill its assigned role (to protect its driver and passengers).

- Reciprocity: “Do not impose on others what you would not desire for yourself.” In general, we do not wish for others to mistreat us in any way. It is reasonable that we should not mistreat others in these ways either.
- Rule 2: an autonomous car should not act in ways that would afflict the highest displeasure or the lowest preference onto other human beings, when other options are available

- Humanity: “A person of humanity is someone who, wishing himself to be established, sees that others are established, and wishing himself to be successful, see that other are successful”.
- Rule 3: an autonomous car must render assistance to other human beings in their pursuit of moral improvement, unless doing so would violate rules 1 and 2; it must also refuse assistance to other human beings when their projects would bring out their evil qualities or produce immorality.

GENERAL CONFUCIAN VIRTUES

- Filial piety (xiao): care for your parents and ancestors
- Kindness (ci): care for your children and descendants
- Righteousness (yi): treat everyone appropriately
- Fidelity (xin): be honest and sincere to everyone
- Humanity/benevolence (ren): universal + differentiated

ALTERNATIVE CONFUCIAN AUTONOMOUS CAR?

- Prof. Fan: virtue-based, relation-relevant, wisdom-associated, and context-sensitive
    1. It should always have a steering wheel available for a human driver to take over and control at any time as he/she wishes;
    2. Those for special public purposes (such as school bus, or elderly home bus) should be programmed to protect the passengers;
    3. Private autonomous cars should be programmed as deontological ones (i.e., in a crisis it is “always to steer for a solid barrier rather than to select human targets”).

LEGAL CHALLENGES

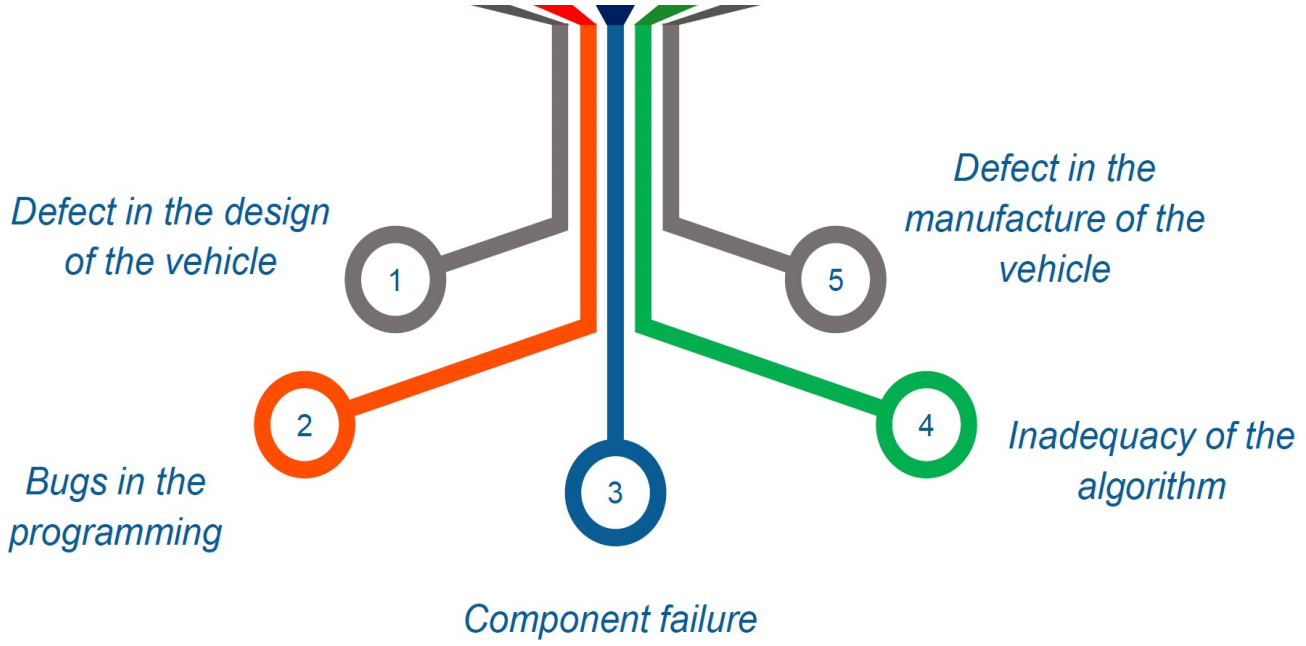
1. Legal liability for car accident
2. Data protection

TESLA (AUTOPILOT) ROAD TRAFFIC ACCIDENTS

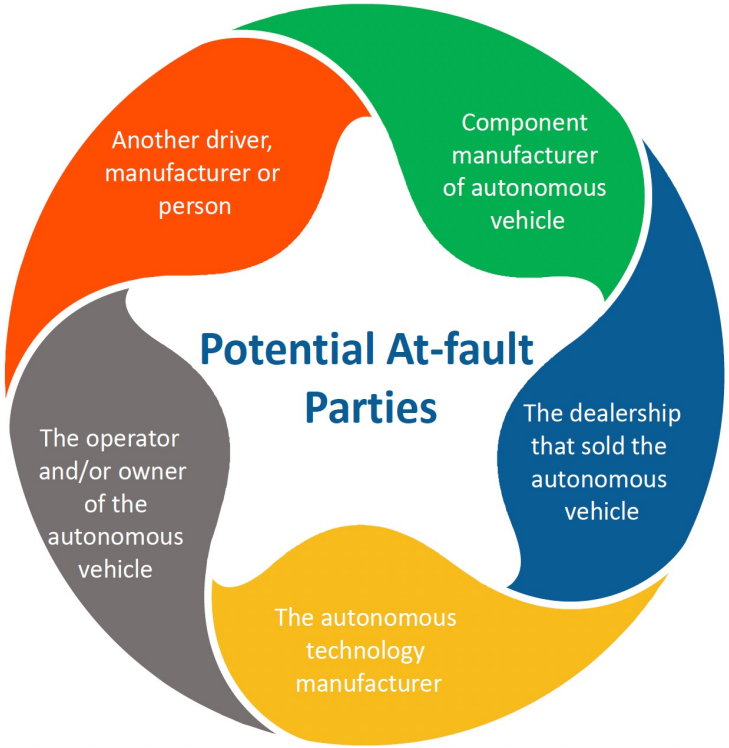
LEGAL LIABILITY FOR CAR ACCIDENT

- Sue the car not sue me?
- Who are potentially liable in law?
- The driver's liability for negligent driving?
- The manufacturer's product liability?
- Who should buy liability insurance?

WHY ACCIDENT CAUSED BY AUTONOMOUS CARS?



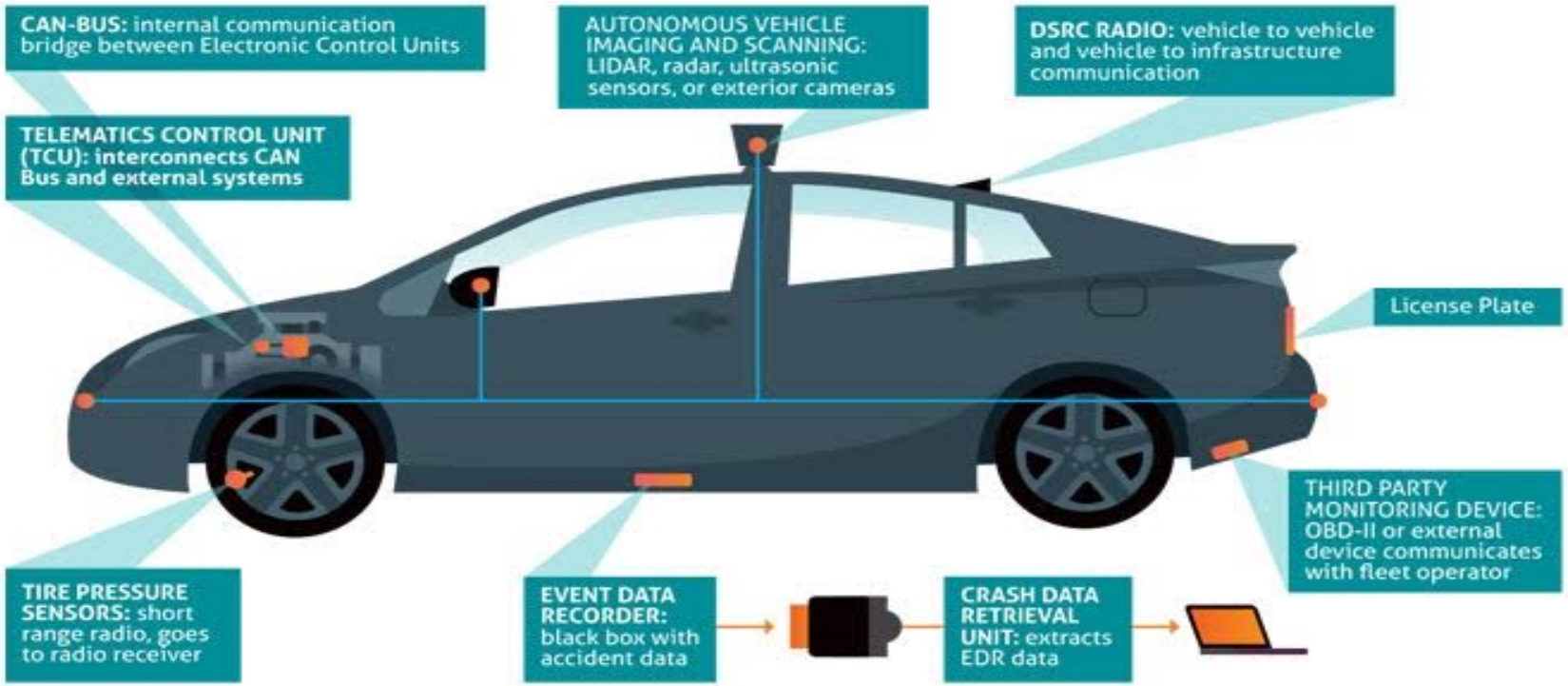
POTENTIAL LIABLE PARTIES



LEGAL CHALLENGES

1. Legal liability for car accident
2. Data protection

DATA and the CONNECTED CAR



CATEGORIES OF AUTOMOBILE DATA

(1) “automobile data”: personal information and important data involved in the process of automobile design, production, sales, use, operation and maintenance, among others;

(2) automobile “personal information”: all kinds of information related to identified or identifiable vehicle owners, drivers, passengers and people outside vehicles, among others, who are electronically or otherwise recorded, but excluding anonymized information;

(3) automobile “sensitive personal information”: the personal information that, once leaked or illegally used, may cause discrimination against vehicle owners, drivers, passengers or people outside vehicles, among others, or serious harm to personal or property safety, including vehicle trajectory, audio, video, images, biometric identification features and other information;

(4) automobile “important data” cover the data that may endanger national security, public interests or the lawful rights and interests of individuals or organizations once it has been tampered with, destroyed, leaked, or illegally obtained or used.

(a) geographical information, personnel flow, vehicle flow and other data of important sensitive areas, such as military management zones, national defense science and industry entities, and Party and government offices at or above the county level;

(b) data reflecting economic operations such as vehicle flow and logistics;

(c) operating data of vehicle charging networks;

(d) video and image data outside vehicles, including face information and license plate information, among others;

(e) personal information involving more than 100,000 personal information subjects.

DATA PROTECTION CONCERNS

- Utility of automobile data: collection, storage, process, use, disclosure…
- Dimension of personal data protection
- Dimension of National security
- Cross-border data flows